

assignment-02-code

August 28, 2026

1 Assignment II

1.1 Import External Libraries

```
[1]: import numpy as np
import sympy as sp

from IPython.display import display, Math
from sklearn.preprocessing import normalize

[2]: BRAKET_O1_BASIS = "\\vert 0 \\rangle / \\vert 1 \\rangle"
BRAKET_PM_BASIS = "\\vert + \\rangle / \\vert - \\rangle"
```

1.2 2. Normalization Using Python

```
[3]: v1 = np.array([[2], [1]])
display(Math(f"\\text{{Original Vector: }} {sp.latex(sp.Matrix(v1))}"))

v2 = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"\\text{{Normalized Vector: }} {sp.latex(sp.Matrix(v2))}"))
```

Original Vector: $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Normalized Vector: $\begin{bmatrix} 0.8944 \\ 0.4472 \end{bmatrix}$

1.3 4. Normalization Using Python with Complex Coefficient

```
[4]: v1 = np.array([[2], [1+1j]])
v2 = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"\\text{{Normalized Vector: }} {sp.latex(sp.Matrix(v2))}"))
```

Normalized Vector: $\begin{bmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{bmatrix}$

1.4 7. Vector Normalization by Hand and Using Python

```
[5]: # v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = np.round(M @ v1_01, 4)
display(Math(f"v_1 \\\text{{ in }} {BRACKET_PM_BASIS} \\\text{{ basis_\\rightarrow(\\text{{ unnormalized}})}}"))
display(Math(sp.latex(sp.Matrix(v1_pm.ravel()))))

print("-----")

v1_pm_normalized = np.round(v1_pm / np.linalg.norm(v1_pm), 4)
display(Math(f"v_1 \\\text{{ in }} {BRACKET_PM_BASIS} \\\text{{ basis (normalized):\\rightarrow}}"))
display(Math(sp.latex(sp.Matrix(v1_pm_normalized.ravel()))))
```

v_1 in $|+\rangle/|-\rangle$ basis (unnormalized):

$$\begin{bmatrix} 2.1213 \\ -0.7071 \end{bmatrix}$$

v_1 in $|+\rangle/|-\rangle$ basis (normalized):

$$\begin{bmatrix} 0.9487 \\ -0.3162 \end{bmatrix}$$

1.5 9. Vector Normalization by Hand and Using Python 2

```
[6]: v1 = np.array([[1], [2]])
v1_normalized = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"v_1 \\\text{{ in }} {BRACKET_O1_BASIS} \\\text{{ basis (normalized):\\rightarrow}}"))
display(Math(sp.latex(sp.Matrix(v1_normalized.ravel()))))

print("-----")

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
display(Math(f"P(\\vert 0 \\rangle) = {prob_0:.4f}"))
```

v_1 in $|0\rangle/|1\rangle$ basis (normalized):

$$\begin{bmatrix} 0.4472 \\ 0.8944 \end{bmatrix}$$

$$P(|0\rangle) = 0.2000$$
